

Week 1: (July 5th - July 11th)

TASKS: Task List

Meeting 1 (7/5: Sunday 10am-12pm)

- Github Repository basic setup: [link](#)
- IBDDMI - Team 7 Google Folder setup:  IBDDMI - Team 7
-  Teamwork Agreement Template.docx

To-Dos

- ☒ ~~Setup GitHub Project Repository~~
- ☒ ~~Complete Teamwork Agreement Document~~
- ☒ ~~Assign Tasks~~

Meeting 2 (7/6: Monday 10am-12pm)

- Reviewed recommended databases
- Chosen Data Type: MPDD
 - Good Subdirectories
 - Needed to develop naming convention for consistency and efficiency
- Tested Collaborating methods (Matlab onedrive, Github)
- Decided to commit for Github for collaboration
- Uploaded MPDD to github

To-Dos

- ☒ ~~Review the chosen dataset (MPDD) and naming scheme (Dayana)~~
- ☒ ~~Ensure all documentation is up to date and everyone is on the same page as to what is next~~
- ☒ ~~Start on greyscale images (?)~~

Meeting 3 (7/8: Wednesday 2-3pm)

- Use imagedatastore or datastore
- Find a way to rename images that have the same file name.
- Assigned tasks for the remainder of the week and updated the task list accordingly.

To-Dos

- ☒ ~~Experiment on filtering images~~
- ☒ ~~Experiment on extracting data to identify scratches, indents, etc.~~
- ☒ ~~Filter images to extract data.~~

Week 2: (July 12th - July 18th)

Meeting 3 (7/12:Sunday 1pm-2pm)

IMPORTANT NOTES FROM MEETING:

- Recap of changes made
RGB (red, green, blue) -> HSV (hue, saturation, value) -> Grayscale (gs) -> Median filter (removing noise) -> Adaptive threshold (helps with uneven lighting) -> Hole filling -> Small object removal -> Morphological opening (so matlab doesn't think two tubes are 1 tube when they kinda touch) -> regionprops (Area, Centroid, Major Axis Length, Minor Axis Length, Orientation, Eccentricity – measures objects to help with classification)
- Focus on step 3: Integrating an AI Classifier into our Single-Image Inspection Function
^ NOTE: matlab recommends resnet18 but **specifically the imagePretrainedNetwork function and specify the “resnet18” model**

Other

- Assigned tasks for upcoming week and updated task list accordingly.

To-Dos

- ☒ Go over ResNet18 (~~resnet18~~ [\(Not recommended\) ResNet-18 convolutional neural network – MATLAB](#)) or alternative
- ☒ Start implementing ResNet18 (or alternative) into the current image classification function/script in order to obtain classification matrices for entire datasets (i.e. Tube sub-dataset)
- ☒ Improve runtime performance and/or memory utilization of single image classification model.
- ☒ Address issues with classification of tubes. (see bullet points below)
 - ☒ Case 1: Multiple tubes being classified as one tube if they are close together (i.e. touching)
 - ☒ Case 2: Corners of the mask sometimes being classified as a tube (i.e. a BoundingBox rectangle is added on to the original image)
 - ☒ Case 3: Tube is not classified at all if another anomaly is spotted in the same image (i.e. image #1 and image #21 of Tube sub-dataset when selected in the imageBasedDefectDetectionMain.mlx script)
 - ☒ Case 4: Tube is not classified at all (i.e. image #15 of Tube sub-dataset when selected in main script)
 - ☒ Case 5: Anomalous tubes are classified as “good” tubes (i.e. image #35 of tube sub-dataset when selected in main script)

Meeting 4 (7/15: Wednesday 2pm-3pm)

- Created training method for the AI classifier
 - Taking in data set for the ID function and shows a training where it would run that function for a specific period of time
 - Would create a .net file that has the AI classifier
 - Shows a graph where the models is trained multiple times with accuracy from 0-100
- Made changes to single image function where the classifier is included within the function
- Added the additional Ai classifier showing pass/fail and confidence score
- Will make that prediction
- Displayinspectionout to format data output
 - Total numbers of tubes and colors anomaly
- Will show the metrics to show that
- Create structure of dataset
 - Took all pictures from anomalies into one path.
- Will delete other transingscriptfunction

To-Dos

- ☒ ~~Determine time needed to fully implement resnet18~~
- ☒ ~~Go over changes made to the classical inspection function~~
- ☒ ~~Assign tasks to be done before next meeting~~
- ☒ ~~Delete other trainingscript() function~~
- ☐ Edit script to show both indents and colors
 - OR -
- ☒ ~~Model will choose one defect over the other~~
- ☒ ~~Attending office hours for the image editor and expanding parameter issue (Thursday, 5pm-6pm)~~
- ☒ ~~Choosing a classifier~~
- ☒ ~~Change black text with white background~~
- ☒ ~~Try to get classical identifier to detect individual and short tubes~~
- ☒ ~~Maybe convert the inspection file to a regular script?~~
- ☒ ~~Improve classical inspection function~~
- ☒ ~~Question about logic in main.mlx -> Evaluating the OR operator, not searching for two things excludeMask = contains(ds.Files, "gt*" | "mask");~~

Finding files that contain gt* OR contain mask, this one seems to cause a chain reaction where it decreases the confidence interval since this changes which images are in our datastore

excludeMask = contains(ds.Files, "gt") | contains(ds.Files, "mask");

Week 3: (July 19 - July 25th)

Meeting 5 (7/19: Sunday 10am-12pm)

- Ryan explained changes he made to the beginning of batch processing.
- Discussed changes needed for the next meeting and updated task list accordingly.

To-Dos

- ☒ ~~Discuss changes~~
- ☒ ~~Discuss needed changes before next meeting~~

Meeting 6 (7/22: Wednesday 2-3pm)

- Isaac discussed the changes he made to the model
- Discussed changes needed for the next meeting and updated task list accordingly.

To-Dos

- ☒ ~~Discuss changes made to training (Isaac)~~
- ☒ ~~Discuss Batch testing and new 'main' function~~
- ☒ ~~Discuss changes made to report template~~
- ☒ ~~Discuss tasks for the end of the week.~~

Week 4: (July 26th - August 1st)

Meeting 7 (7/26: Sunday 10am-12pm)

- Edited lab report template.
- Assigned roles for lab report.
- Continued working on a robustness test model.

To-Dos

- ☒ ~~Discuss folder names changes (Issac)~~
- ☒ ~~Attend 7/28 meeting (Kaitlyn)~~
- ☒ ~~Assign final tasks for the group~~
 - Finish the single use script (Ryan)
 - Readme in matlab (Ryan)
 - Lab report document
 - Dayana Sections 1-2
 - Kaitlyn sections 3-4
 - Isaac sections 5-6

Meeting 8 (7/29: Wednesday 2pm-3pm)

- Assigned remaining sections of the final technical report to team members.
- Cleared up areas of confusion regarding individual sections in the final technical report.
- Planned out final changes prior to project conclusion and handover/submission.

To-Dos

- ☒ ~~Discuss matlab changes (Ryan & Isaac)~~
- ☒ ~~Discuss report edits (Isaac, Kaitlyn, & Dayana)~~
- ☒ ~~Discuss office hour notes and needed changes for submission (Kaitlyn)~~
- ☒ ~~Assign tasks to be done before the next meeting.~~

Week 5: (August 2nd - August 5th)

Meeting 9 (8/2: Sunday 10am-12pm)

Discussed changes, and changes we believed needed to be made before the next meeting.
Discussed how we would submit the project.
Assigned tasks for upcoming week.

To-Dos

- ☒ Discuss changes made
 - ☒ Ryan (Conclusions & classical inspection)
 - ☒ Dayana (References)
 - ☒ Kaitlyn (Section 3 & 4 of report)
 - ☒ Isaac (Abstract, and model training sections)
- ☒ Discussion of project roles
- ☒ Discuss final tasks
- ☒ Upload Script as PDF in appendix
- ☒ Proofread references
 - ☒ Dyana sections 7 and 8
 - ☒ Kaitlyn sections 5 and 6
 - ☒ Issac sections 3 and 4
 - ☒ Ryan sections 1, 2 and abstract
- ☒ Formatting report
- ☒ Verifying steps

Meeting 10 (8/5: Wednesday 2pm-3pm)

Discussed what we accomplished from the previous meeting.
Double checks all of our files prior to submission.
Uploaded the meeting notes and task list files along with the final report to Github.
Submitted the project for review.

To-Dos

- ☒ Upload lab report, meeting notes, and task list as documents to the github
- ☒ Final Checks and submission
- ☒ Ensure everyone is on the same page about the project going forward